

chap_pipps_ tous les PLANS

2025-07-09

Plan à probabilités inégales et avec remise

```
library(sampling)
data(rec99)
attach(rec99)
(N <- length(LOG))

## [1] 554
# [1] 554
tLOG <- sum(LOG)
pk <- LOG/tLOG

m<-70
set.seed(12345)
(ech.pps.o <- sample(N,m,replace=TRUE,prob=pk))

## [1] 435 220 352 86 544 73 544 337 506 393 31 528 62 44 537 166 145 31 27
## [20] 549 172 525 504 546 234 79 154 205 535 482 31 106 213 269 549 243 544 449
## [39] 432 31 546 167 249 189 324 211 543 392 419 187 512 356 430 260 535 363 491
## [58] 63 244 457 39 468 363 10 77 338 129 145 338 78

(R_ech.pps.o <- table(ech.pps.o))

## ech.pps.o
## 10 27 31 39 44 62 63 73 77 78 79 86 106 129 145 154 166 167 172 187
## 1 1 4 1 1 1 1 1 1 1 1 1 1 1 2 1 1 1 1 1
## 189 205 211 213 220 234 243 244 249 260 269 324 337 338 352 356 363 392 393 419
## 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 2 1 1 1
## 430 432 435 449 457 468 482 491 504 506 512 525 528 535 537 543 544 546 549
## 1 1 1 1 1 1 1 1 1 1 1 1 1 2 1 1 3 2 2

m <- 70
set.seed(12345)
R <- UPMultinomial(m*pk)

table(R)

## R
## 0 1 2 3
## 494 52 6 2

sum(R)

## [1] 70
# [1] 70
(ech.pps.o <- rep((1:N)[R!=0],R[R!=0]))
```

```
## [1] 4 10 21 23 31 39 39 44 52 63 65 70 77 98 103 106 106 108 117
## [20] 133 135 137 146 147 167 167 167 169 171 171 173 178 186 195 227 235 237 241
## [39] 243 261 261 269 269 269 274 337 348 348 353 378 382 389 389 392 393 419 426
## [58] 427 446 449 455 457 488 491 493 506 536 546 549 550
```

```
ech.pps <- which(R!=0)
length(ech.pps)
```

```
## [1] 60
```

```
# [1] 60
```

```
LOGVAC.ech.pps <- (LOGVAC/pk)[ech.pps]
```

```
R_ech.pps <- R[ech.pps]
(LOGVAC.HH <- sum(R_ech.pps*LOGVAC.ech.pps)/m)
```

```
## [1] 10272.93
```

```
# [1] 10272.93
```

```
(var.est.HH <- sum(R_ech.pps*(LOGVAC.ech.pps-LOGVAC.HH)^2)/(m*(m-1)))
```

```
## [1] 565951
```

```
# [1] 565951
```

```
c(LOGVAC.HH-qnorm(0.975)*sqrt(var.est.HH),
  LOGVAC.HH+qnorm(0.975)*sqrt(var.est.HH))
```

```
## [1] 8798.453 11747.403
```

```
# [1] 8798.453 11747.403
```

```
sum(LOGVAC)
```

```
## [1] 10768
```

```
# [1] 10768
```

```
library(survey)
```

```
## Loading required package: grid
```

```
## Loading required package: Matrix
```

```
## Loading required package: survival
```

```
##
```

```
## Attaching package: 'survival'
```

```
## The following objects are masked from 'package:sampling':
```

```
##
```

```
## cluster, strata
```

```
##
```

```
## Attaching package: 'survey'
```

```
## The following object is masked from 'package:graphics':
```

```
##
```

```
## dotchart
```

```
recs <- cbind.data.frame(rec99, poids=1/(m*pk))
recs.o <- recs[ech.pps.o,]
design.pps <- svydesign(id=~1,weights=recs.o$poids, data=recs.o)
```

```
res.pps <- svytotal(~LOGVAC, design.pps)
print(ftable(res.pps), digits=5)
```

```
##                LOGVAC
## LOGVAC 10272.9    752.3
```

```
#LOGVAC
#LOGVAC 10272.9 752.3
```

Plan à probabilités inégales et sans remise

```
n <- 70
pik <- inclusionprobabilities(LOG,n)
```

```
rec99[pik==1,c(2,5)]
```

```
##          COMMUNE  LOG
## 32          AUTERIVE 2777
## 39  BAGNERES-DE- 4490
## 106 CASTELGINEST 2874
## 403 PORTET-SUR-G 3473
## 419          REVEL 3894
## 455  SAINT-JEAN 3288
## 549 VILLENEUVE-T 3189
```

```
# COMMUNE LOG 32 AUTERIVE 2777 39 BAGNERES-DE- 4490 106 CASTELGINEST 2874 403 PORTET-SUR-G 3473 419 REV.
```

```
summary(pik[pik<1])
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.     Max.
## 0.004362 0.023626 0.048705 0.115174 0.128123 0.903224
```

```
#Min. 1st Qu. Median
#Mean 3rd Qu.
#Max.
#0.004362 0.023626 0.048705 0.115174 0.128123 0.90322
```

PLAN POISSON

```
n <- 70
pik <- inclusionprobabilities(LOG,n)
```

```
set.seed(12345)
s <- UPpoisson(pik)
sum(s)
```

```
## [1] 64
```

```
# [1] 64
which(s==1)
```

```
## [1] 11 14 19 21 32 39 48 49 62 73 83 84 85 99 106 119 123 131 145
## [20] 150 154 172 187 189 196 208 211 213 220 229 251 253 257 266 271 278 287 300
## [39] 314 328 329 362 363 390 400 403 414 419 422 423 428 435 437 443 448 455 465
## [58] 482 512 537 544 546 547 549
```

```

set.seed(12345)
ns <- replicate(10000,
{ss<-UPpoisson(pik)
  sum(ss)})
mean(ns)

```

```
## [1] 69.9674
```

```
#[1] 69.9674
```

```

ys <- LOGVAC[s==1]
piks <- pik[s==1]
(LOGVAC.HT <- HTestimator(ys,piks))

```

```
##           [,1]
```

```
## [1,] 9148.435
```

```
# [,1]
```

```
# [1,] 9148.435
```

```
(var.est.HT <- sum((1-piks)*(ys^2)/(piks^2)))
```

```
## [1] 1037840
```

```
#[1] 1037840
```

```

pikl <- outer(pik,pik,"*")
diag(pikl) <- pik
pikls <- pikl[s==1,s==1]
varHT(ys,pikls,method=1)

```

```
## [1] 1037840
```

```
#[1] 1037840
```

```

c(LOGVAC.HT-qnorm(0.975)*sqrt(var.est.HT),
  LOGVAC.HT+qnorm(0.975)*sqrt(var.est.HT))

```

```
## [1] 7151.732 11145.137
```

```
#[1] 7151.732 11145.137
```

```
sum(LOGVAC)
```

```
## [1] 10768
```

```
#[1] 10768
```

Plan systématique à probabilités inégales

```

n <-70
set.seed(12345)
s <-UPsystematic(pik)
sum(s)

```

```
## [1] 70
```

```
#[1] 70
```

```

pikl <- UPsystematicpi2(pik)
sum(pikl==0)

```

```
## [1] 231304
# [1] 231304
```

```
piks <- pik[s==1]
pikls <- pikl[s==1,s==1]
```

```
ys <- LOGVAC[s==1]
HTestimator(ys,piks)
```

```
##           [,1]
## [1,] 10638.29
# [1,] 10638.29
```

```
varHT(ys,pikls,method=2)
```

```
## [1] -15287556
# [1] -15287556
```

```
varest(ys,pik=piks)
```

```
## [1] 393759.6
# [1] 393759.6
```

Plan de Poisson conditionnel à la taille

```
set.seed(12345)
s <- UPmaxentropy(pik)
sum(s)
```

```
## [1] 70
# [1] 70
```

```
piktilde <- pik
piktilde[pik!=1] <- UPMEpiktildefrompik(pik[pik!=1])
```

```
piktildes <- piktilde[s==1]
pikltilde2 <- UPmaxentropypi2(pik)
pikltilde2s <- pikltilde2[s==1,s==1]
```

```
ys <- LOGVAC[s==1]
(LOGVAC.HT <- HTestimator(ys,piktildes))
```

```
##           [,1]
## [1,] 10747.63
# [,1]
# [1,] 10747.63
(V.HT<-varHT(ys,pikltilde2s,method=2))
```

```
## [1] 537803.7
# [1] 537803.7
```

```
c(LOGVAC.HT-qnrm(0.975)*sqrt(V.HT),
  LOGVAC.HT+qnrm(0.975)*sqrt(V.HT))
```

```
## [1] 9310.284 12184.967
```

```
# [1] 9310.284 12184.967
```

```
sum(LOGVAC)
```

```
## [1] 10768
```

```
# [1] 10768
```